Structural-Hypothesis Routing in Synthetic Static Nonlinear Operators: A Selection-Separated Simulation Study with Negative Ablations

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Abstract

Model-family choice is itself a source of uncertainty in nonlinear input-output modeling: linear, polynomial, kernel, latent, and structured nonlinear representations can each be appropriate under different mechanisms. We study a deliberately narrow question in **synthetic static complex-valued operator identification**: can heterogeneous model families be routed effectively when model nomination, independent confirmation, empirical release screening, and final evaluation are assigned separate random data roles?

We evaluate STABLE-Wave MX5, a frozen twelve-candidate structural router, on internally developed randomized operator families. In the primary 24-task frozen cohort, MX5 released 22/24 tasks and matched the retrospective test-best candidate on 24/24 tasks. A second post-freeze 24-task cohort again released 22/24 tasks. Across both cohorts, 0/44 released tasks exceeded the predeclared 10^{-3} test-NMSE operating threshold; however, the corresponding simple one-sided zero-event 95% upper bound is still about 6.6%, so this result is **not** a formal safety or finite-sample risk guarantee. Twelve independently permuted negative controls were rejected.

Two negative ablations materially limit the interpretation. First, on the primary benchmark, full six-way data separation, rank-only routing, and same-split reuse produce identical expert choices, release decisions, and test errors. A separate 12-task structural-boundary stress set yields only one gate-induced structural correction and no release-decision difference. Second, when fixed baselines are given an equal total observation budget, mean errors still favor MX5, but family-level $n = 8$ inference is not significant after Holm correction. Thus the evidence does **not** establish that six hard splits are necessary, nor that MX5 is superior under equal information budgets.

The main result is therefore procedural rather than architectural: **within an internally developed synthetic generator program, heterogeneous structural hypotheses can be routed accurately, while strict data-role separation appears conservative rather than demonstrably necessary when candidate margins are large**. The study does not establish external generalization, temporal dynamical-system identification, same-budget superiority, or formal release-risk control.

Keywords: simulation methods; structural model routing; nonlinear operator identification; complex-valued regression; negative ablation; model selection; selective release; synthetic benchmark.

Scope of Claims and Evidence Levels

The paper should be read primarily as a **simulation-methods and methodological stress-test study**. The strongest contribution is therefore the combination of a positive internal routing result and negative evidence about the necessity and equal-budget superiority of the full protocol. MX5 is the instrument used to study structural routing; it is not presented as a new universal system-identification architecture. The strongest positive result concerns internal routing accuracy. The strongest negative result is that the present experiments do not demonstrate a material benefit from six-way evidence separation or equal-budget superiority over fixed alternatives.

This manuscript separates three evidence levels.

Evidence level	Role	Can it support the paper's primary claim?
Developmental evolution	Architecture discovery, failure analysis, rejected branches, hard-family creation	No; used to design MX5

Evidence level	Role	Can it support the paper's primary claim?
Primary frozen internal validation	24 fresh random tasks nested in 8 transformation families, generated after MX5 and the 10^{-3} release tolerance were frozen	**Yes; primary quantitative evidence within the internal generator program**
Post-freeze validation and challenges	A second 24-task cohort, a 12-task structural-boundary family, new composed mechanisms, input-distribution shifts, expanded negative controls, and sample-efficiency probes	Supportive validation/stress evidence; not used to alter MX5

The primary 24-task benchmark is **frozen internal validation relative to the MX5 implementation**, but it was not prospectively registered with an external registry before the overall research program began. The paper therefore uses the terms *frozen internal validation* and *post-freeze validation* rather than *preregistered* or *external validation*. No architecture, candidate-pool, routing, gate, or release screening-threshold change was made in response to the final 24-task test labels.

The central claim is deliberately bounded:

> **Within the randomized static complex-valued transformation families studied here, selection-separated routing among heterogeneous structural hypotheses can outperform forcing a single tested model family globally, while an independent empirical release screen can abstain on unsupported tasks.**

The manuscript does **not** claim a universal operator-identification procedure, temporal dynamical-system identification, formal post-selection inference, distribution-free risk control, external validation, or demonstrated superiority on measured physical systems.

1. Introduction

The term *operator identification* is used throughout this manuscript to keep the empirical scope precise. The studied mappings are static across samples: they may contain dense, local, low-rank, bilinear, latent, and other nonlinear interactions within a complex-valued vector, but they do not contain hidden temporal state, feedback, or long-memory dynamics. The work is therefore related to nonlinear system identification, but it should not be read as an evaluation on general dynamical-system identification problems.

More broadly, system identification asks a basic but difficult question: given paired observations of a system input and output, what representation should be used to predict the system response on new inputs? Classical identification offers linear, polynomial, Volterra, kernel, state-space, nonlinear autoregressive, and many other representations. Modern machine learning adds deep networks and mixture-of-experts architectures. The practical difficulty is therefore no longer only parameter estimation. It is also **structural choice**: the correct inductive bias for one system may be actively unhelpful for another.

This issue is especially visible in complex-valued multicarrier transformations. A nearly diagonal system with smooth carrier response is well matched by a compact symbolic transfer function. A system with sparse cross-carrier quadratic interactions may favor a second-order polynomial or Volterra-like model. A transformation driven by a small number of unknown global carrier mixtures may instead favor a learned latent-projection model. A universal black-box model can, in principle, approximate many of these mechanisms, but doing so may require more data, more optimization, and less interpretability than choosing an appropriate structural hypothesis.

Mixture-of-experts methods address heterogeneity by combining specialized models [3,4]. However, standard expert gating is usually framed as a learned sample-level or context-level routing problem. STABLE-Wave takes a different approach. The object being routed is the **entire structural hypothesis for an unknown transformation**. For each identification task, multiple model families are fitted and compared. The system then selects one complete identifier for deployment on that transformation.

That formulation introduces a statistical danger. If the same held-out observations are repeatedly used to tune hyperparameters, choose the model family, decide whether the chosen family improves on a fallback, calibrate an acceptance rule, and report final performance, the resulting evaluation becomes progressively less independent. Recent work on routing diagnostics has highlighted the broader danger of reusing evaluation data after selecting the best candidate from a pool [9]. Similar concerns arise throughout empirical model selection.

We therefore ask the following research question:

> **Can heterogeneous nonlinear transformations be identified more robustly by treating model families as competing structural hypotheses, assigning distinct random datasets to tuning, ranking, confirmation, risk screening, and final evaluation, and abstaining when the evidence does not support a predeclared error target?**

1.1 Frozen Internal-Validation Questions

Before interpreting the final 24-task benchmark, four internal-validation hypotheses define what would count as success.

H1 — Routing utility under the original evaluation budget. The adaptive MX5 procedure should have lower family-level test NMSE than each prespecified fixed baseline: dense complex linear regression, RBF kernel ridge regression, and polynomial kernel ridge regression. A separate equal-total-observation-budget analysis is treated as a post-hoc fairness diagnostic rather than part of H1.

Q2 — Empirical release behavior. Among tasks released by the independent release-screen split, the observed test NMSE should remain below the predeclared 10^{-3} operating threshold.

Q3 — Signal dependence. When the input-output pairing is independently destroyed in every label-bearing pre-test split, MX5 should refuse release rather than manufacture a low-error mapping from marginal statistics alone.

Q4 — Structural specialization. The router should select more than one expert family across heterogeneous tasks; a single candidate dominating every family would weaken the structural-routing interpretation.

These questions separate performance, selective reliability, negative-control behavior, and structural diversity. Q1 is evaluated primarily with paired **family-level** tests and multiplicity correction; task-level tests are retained only as sensitivity analyses because tasks are nested within family. Q2 is evaluated as an observed threshold-violation count together with exact binomial uncertainty; it is an empirical operating-behavior hypothesis, not a formal safety guarantee. Q3 and Q4 are descriptive stress hypotheses rather than claims of formal statistical power.

STABLE-Wave MX5 is an empirical answer to that question. It uses six disjoint random input-output splits for each task:

$$D_{\text{train}},\ D_{\text{tune}},\ D_{\text{rank}},\ D_{\text{gate}},\ D_{\text{cert}},\ D_{\text{test}}.$$

The first two fit and tune the candidate models. The rank split nominates a structural hypothesis. The gate split independently tests whether the nominee improves on a simpler fallback. The release-screen split estimates whether the selected identifier is sufficiently accurate to release. The test split is never used until all decisions are frozen.

The main contributions of this work are:

1. **A controlled structural-routing benchmark.** Twelve heterogeneous model families compete to explain each static complex-valued transformation.
2. **A strict evidence-separation protocol.** Train, tune, rank, gate, empirical release-screen, and test roles are separated by construction.
3. **A full forced-expert diagnostic.** Every frozen candidate is evaluated on every primary test task, allowing routing regret to be measured directly.
4. **Negative ablations of the protocol itself.** Rank-only routing, same-split reuse, equal-budget baselines, and a structural-boundary stress family test whether the elaborate protocol is actually necessary.
5. **Transparent evidence boundaries.** Developmental evolution, frozen internal validation, and post-freeze stress evidence are reported separately.
6. **Explicit negative conclusions.** The study reports that six-way splitting is not shown to improve release behavior on the present benchmark and that same-budget superiority is not established after family-level multiplicity correction.

The claims are deliberately narrower than “mixture-of-experts is new” or “selective prediction is new.” Both have extensive prior art [3–8]. The contribution evaluated here is the combination of heterogeneous **structural** hypotheses with a selection-separated evidence pipeline for whole-transformation routing.

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2. Related Work

2.1 Nonlinear system identification

System identification has a long history of choosing model structure according to expected system dynamics [1]. Polynomial and Volterra representations remain important because they provide systematic expansions for nonlinear input-output relations. Recent work has connected Volterra representations with regularized Hilbert-space and kernel formulations, preserving classical structure while using modern statistical learning machinery [2].

The present work is related to that tradition but addresses a different decision layer. Rather than committing to one nonlinear representation, MX5 maintains a finite pool containing linear, structured residual, latent nonlinear, and kernel models. The routing decision is therefore itself part of the identification procedure.

Piga and Forgione’s recent ASIA framework also emphasizes that practical identification involves iterative choices over model class, architecture, training strategy, and hyperparameters [10]. ASIA delegates much of that search to an autonomous coding agent. MX5 instead defines a fixed, reproducible candidate family and focuses on data-separated statistical selection among those structural hypotheses.

2.2 Mixture of experts and expert routing

Hierarchical mixtures of experts established the general principle that specialized models can be combined through learned gates [3]. More recently, Rajabi and Salehi applied context-aware gated mixtures of LSTM experts to nonlinear and nonstationary system identification, showing improved robustness under regime changes [4]. These works demonstrate that expert specialization and routing are established ideas.

MX5 differs in three practical respects. First, routing occurs once per identified transformation rather than at every sample. Second, the candidate experts need not share an architecture or even a learning paradigm: dense ridge models, topology-aware semi-parametric regressors, latent neural residual models, and kernel regressors can compete in the same pool. Third, the routing evidence is split across rank and gate datasets so that the model that looks best on the nomination set must still survive independent confirmation.

2.3 Selective prediction and risk control

Selective prediction permits a model to abstain when confidence is insufficient. Deep selective classifiers and SelectiveNet demonstrated risk-coverage trade-offs in modern neural prediction [5,6]. More recent selective conformal work provides explicit finite-sample or PAC-style risk-control mechanisms under stated assumptions [7,8].

MX5 adopts the operational idea of selective release but **does not claim conformal coverage**. Its release-screen stage uses a nonparametric bootstrap distribution of release screening-split mean normalized errors and compares an empirical upper percentile against a fixed tolerance. This provides a practical risk screen in the experiments reported here, but it is not a distribution-free guarantee. A conformal or e-value replacement for this stage is a natural theoretical extension.

2.4 Selection after choosing among candidates

Routing and model selection can create optimistic evaluation when the best model or policy is chosen on the same examples used for inference. Shihab et al. recently formalized this issue in multi-model routing and derived selection-valid diagnostics that separate oracle opportunity from deployable routing performance [9]. Although their domain is language-model routing rather than system identification, the statistical lesson is directly relevant: **candidate selection and router evaluation should not be silently conflated**.

MX5 addresses this procedurally through disjoint data roles. This study does not present a theorem equivalent to the guarantees in [9]; instead, it evaluates whether strict separation of nomination, confirmation, release screening, and test sets produces reliable empirical routing in randomized nonlinear identification.

2.5 Novelty Boundary

The paper therefore makes a **methodological-procedural claim**, not a model-architecture claim. A reviewer should be able to remove the name "STABLE-Wave" from the manuscript and still understand the scientific question: whether heterogeneous structural hypotheses can be routed reliably under separated evidence roles, and whether that separation materially changes outcomes. The negative ablations are part of the contribution because they show that the second part is not established on the present benchmark.

The individual ingredients of MX5 are not claimed as new in isolation. Mixtures of experts date back decades, nonlinear system identification has extensive linear, Volterra, kernel, neural, and grey-box traditions, and selective prediction/risk control is an established research area. Recent 2026 work specifically applies context-aware mixtures of LSTM experts to nonlinear system identification, while contemporary routing methodology emphasizes the distinction between oracle opportunity and selection-valid deployable routing.

Accordingly, the proposed contribution is **not** any of the following:

- the existence of multiple experts;
- polynomial or RBF kernel regression;
- latent nonlinear features;
- abstention;
- bootstrap resampling;

- or the general idea of model selection.

The contribution evaluated here is the **procedure-level composition**: complete structural model families are treated as competing hypotheses for an entire unknown transformation; internal tuning is separated from structural nomination; a nominated structure must survive independent paired gate evidence against a simpler rank-selected fallback; release is then screened on a separate release-screen sample; and final error is measured only after those decisions are frozen.

This distinction matters because a heterogeneous expert pool can show a large retrospective oracle gap even when no deployable router can realize it. MX5 therefore reports deployable frozen routing performance rather than only the performance of the best test-set expert.

3. Static Complex-Valued Operator Identification Problem

For a fixed task, let

$$x_i \in \mathbb{C}^K, \qquad y_i = T(x_i) \in \mathbb{C}^K,$$

where T is an unknown complex-valued transformation and K is the number of carriers or complex coordinates. The transformations studied here are **memoryless across samples** but can contain dense, local, low-rank, nonlinear, and cross-coordinate interactions within each K-dimensional sample. The present paper therefore does not claim to solve temporal dynamical-system identification.

Given disjoint random samples from the same task transformation, the objective is to learn a predictor

$$\hat{T} : \mathbb{C}^K \to \mathbb{C}^K$$

with low normalized mean-squared error (NMSE),

$$\mathrm{NMSE}(\hat{Y}, Y) = \frac{\mathbb{E}\|\hat{Y} - Y\|_2^2}{\mathbb{E}\|Y\|_2^2 + \epsilon}.$$

The selective version of the problem also requires an acceptance decision

$$A \in \{0, 1\},$$

where $A = 1$ means that the fitted transformation is considered supported for release. MX5 uses the operational tolerance

$$\tau = 10^{-3}.$$

The desired empirical behavior is therefore twofold:

1. achieve low NMSE across heterogeneous transformation mechanisms; and
2. when independent evidence suggests that the target error is not met, abstain rather than force a prediction.

3.1 Primary Estimands

The paper reports four primary quantities.

1. **Raw routing error:** mean task-level test NMSE over all 24 frozen tasks, regardless of whether the task is eventually released.
2. **Selective coverage:** the proportion of tasks that pass the independent release-screen rule.
3. **Selective risk:** test NMSE among released tasks, summarized by mean, median, maximum, and threshold-violation count.
4. **Routing gain:** paired task-level error difference between MX5 and each fixed external baseline.

This separation prevents a common ambiguity in selective systems. Low error can be obtained trivially by refusing almost everything, while high coverage can be obtained trivially by releasing everything. MX5 is therefore evaluated on **risk and coverage jointly**, with the 10^{-3} target fixed before final evaluation.

4. STABLE-Wave MX5

4.1 Six disjoint data roles

Each randomized task produces six independently sampled input-output datasets from the same frozen transformation:

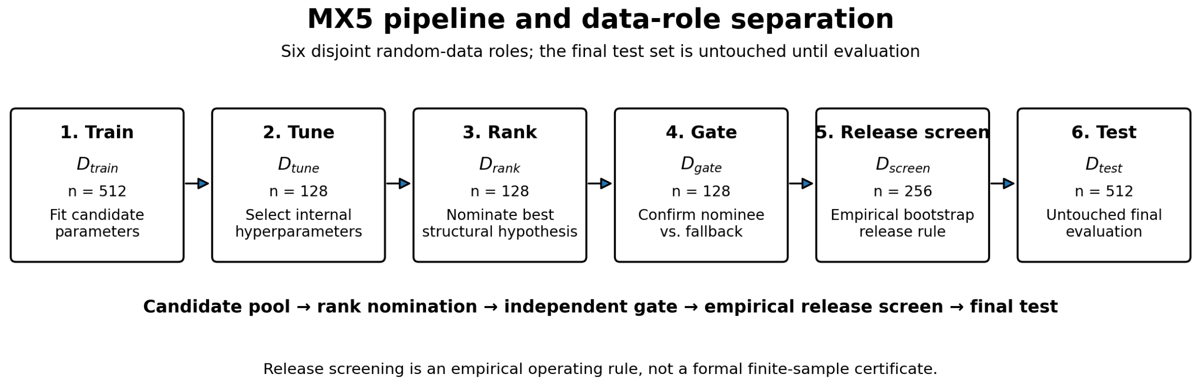

Figure 1: MX5 pipeline and disjoint data roles. Each random-data split has a distinct role, and the final test set is untouched until scientific evaluation.

Split	Size in final benchmark	Purpose
Train	512	Estimate model parameters
Tune	128	Select internal hyperparameters
Rank	128	Nominate a structural candidate
Gate	128	Independently confirm nominee vs. fallback
Release screen	256	Apply the empirical release rule after routing is fixed
Test	512	Final untouched evaluation

The task transformation itself is common across these splits, but the input samples are independently generated. No test target is used for fitting, tuning, routing, gating, or release screening.

4.2 Symbolic complex carrier response

Several MX5 candidates begin with an interpretable diagonal carrier transform. For carrier k, an initial complex response estimate is

$$r_k = \frac{\sum_i \overline{x_{ik}} y_{ik}}{\sum_i |x_{ik}|^2 + \epsilon}.$$

Its gain and unwrapped phase are modeled separately. The carrier coordinate is mapped to a compact basis containing a constant term, polynomial terms through degree three, four sine/cosine harmonics, and boundary indicators. Ridge regression fits gain in decibels and unwrapped phase:

$$g_k = b_k^\top \beta_g, \qquad \phi_k = b_k^\top \beta_\phi.$$

The resulting symbolic mask is

$$m_k = 10^{g_k/20} e^{j\phi_k}.$$

This stage captures smooth global carrier structure with a compact, inspectable representation.

4.3 Structured and dense complex linear bridges

After the symbolic mask,

$$Z = X\,\mathrm{diag}(m),$$

MX5 considers two linear backbones.

The **dense bridge** uses complex ridge regression,

$$W_d = (Z^H Z + \alpha I)^{-1} Z^H Y.$$

The **structured bridge** first fits a dense solution, then decomposes the deviation from identity into a circular banded component and a low-rank long-range component:

$$W_s = I + B_{\text{band}} + U_r \Sigma_r V_r^H.$$

The frozen configuration uses circular bandwidth two and rank two, followed by singular-value clipping for stability. The structured bridge therefore represents local cross-carrier coupling plus a small number of global linear modes, while the dense bridge makes fewer structural assumptions.

4.4 Topology-adaptive nonlinear residual expert

For a bridge output Z, the residual target is

$$R = Y - Z.$$

MX5 scores circular carrier offsets according to the correlation of shifted input-derived features with the residual. It always includes a small local neighborhood and then adds the most informative discovered offsets.

For each selected offset, the feature bank includes real and imaginary parts, magnitude-squared terms, hyperbolic-tangent nonlinearities, and cross-products with the center carrier. Per-output ridge regressions map these features to complex residual corrections.

Two topology budgets are fitted: a compact discovered-offset model and a nearly global model. Tune-set error determines whether one or both are retained in an equal-weight ensemble. This produces the `structured_base` and `dense_base` candidates depending on the upstream bridge.

4.5 Shared-context and per-output context experts

Some transformations depend on a latent mixture shared across all outputs; others use output-specific mixtures.

The **shared-context expert** builds a circular shift design matrix from the bridge output, learns a regularized shared context signal, and expands that signal through nonlinear features including saturating functions and energy-sensitive modulation. Those context features are concatenated with the topology-adaptive feature tensor before the final residual ridge fit.

The **per-output expert** learns an output-specific latent context from the full real/imaginary bridge representation. Each output carrier receives its own complex context coordinates, which are expanded using a fixed nonlinear basis before ridge fitting.

These mechanisms yield four candidates:

- `structured_shared`,
- `structured_per_output`,
- `dense_shared`,
- `dense_per_output`.

4.6 Latent-projection nonlinear experts

Hard random transformations revealed a separate failure mode: nonlinear responses driven by a small number of unknown global complex mixtures. MX5 therefore includes a compact latent residual model.

For normalized complex input z, the model learns q complex projections

$$c = zW, \qquad q \in \{10, 14\}.$$

A fixed nonlinear basis is then applied to each latent coordinate. It contains linear real/imaginary components, saturating hyperbolic tangents, rational terms of the form

$$\frac{\Re c}{1 + |c|^2}, \qquad \frac{\Im c}{1 + |c|^2},$$

energy-centered saturation, and odd energy-weighted terms. A learned linear map converts those latent features into a complex residual.

The latent model is fitted at two different points in the pipeline:

- `masked_latent`: immediately after the symbolic diagonal mask;
- `dense_latent`: after the dense complex linear bridge.

The latent dimension $q \in \{10, 14\}$ is selected on the tune split only.

4.7 Polynomial and RBF kernel residual experts

The final two candidates are classical kernel ridge regressors. Both begin from the symbolic-plus-dense-linear baseline and model the remaining complex residual from standardized real/imaginary features.

The polynomial expert searches degree $d \in \{2, 3\}$, additive constant $c \in \{0, 1\}$, and ridge penalties $\alpha \in \{0.01, 0.1, 1\}$.

The RBF expert uses a median-distance scale and searches three kernel-width multipliers and the same ridge-penalty grid.

These candidates are important methodologically. MX5 does not define success as "the new bespoke expert must beat classical methods." If a polynomial model is best supported by the data, the router is expected to select it.

4.8 Frozen Routing Procedure

The complete deployment logic can be summarized as follows.

```
Input:
    D_train, D_tune, D_rank, D_gate, D_screen
    candidate family F
    simple fallback family S
    release tolerance tau = 1e-3

1. Fit every candidate f in F using D_train.
2. Use D_tune only for candidate-internal hyperparameter selection.
3. Evaluate all frozen candidates on D_rank.
4. nominee <- lowest-error candidate on D_rank.
5. fallback <- lowest-error candidate in S on D_rank.
6. If nominee != fallback:
      compare paired per-sample errors on D_gate.
      accept nominee only if the predeclared gate criterion is met;
      otherwise select fallback.
7. Evaluate the selected identifier on D_screen.
8. Bootstrap the mean normalized release screening loss 600 times.
9. supported <- (95th percentile of bootstrap means <= tau).
10. If supported:
      release predictor.
    Else:
      abstain / mark transformation unsupported.
11. D_test is opened only for final scientific evaluation.
```

The rank split is allowed to search the candidate pool; the gate split is not. The gate answers only a binary post-nomination question: whether the nominated structure shows independent evidence of improvement over a simpler fallback. Release screening occurs after the routing choice is fixed.

4.9 Candidate pool

The frozen MX5 candidate pool contains 12 complete structural hypotheses:

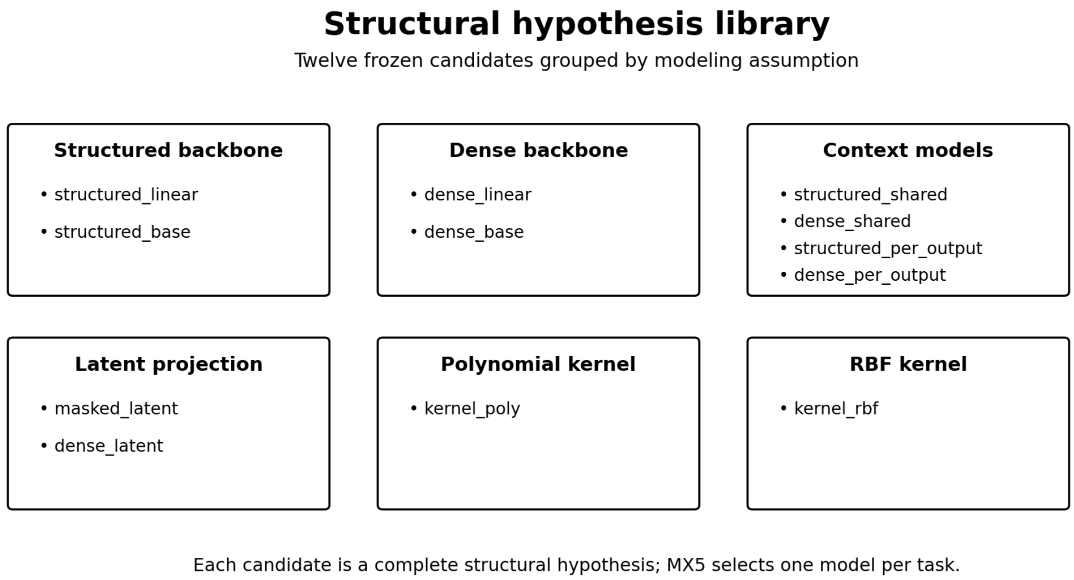

Figure 2: Frozen structural-hypothesis library. The twelve candidates span structured, dense, contextual, latent-projection, polynomial-kernel, and RBF-kernel assumptions.

Candidate	Bridge / representation	Residual model
`structured_linear`	Symbolic + structured bridge	None
`structured_base`	Symbolic + structured bridge	Topology-adaptive ridge ensemble
`structured_shared`	Symbolic + structured bridge	Shared-context nonlinear ridge
`structured_per_output`	Symbolic + structured bridge	Per-output context nonlinear ridge
`dense_linear`	Symbolic + dense bridge	None
`dense_base`	Symbolic + dense bridge	Topology-adaptive ridge ensemble
`dense_shared`	Symbolic + dense bridge	Shared-context nonlinear ridge
`dense_per_output`	Symbolic + dense bridge	Per-output context nonlinear ridge
`masked_latent`	Symbolic mask	Latent-projection nonlinear residual
`dense_latent`	Symbolic + dense bridge	Latent-projection nonlinear residual
`kernel_poly`	Symbolic + dense linear	Polynomial KRR residual
`kernel_rbf`	Symbolic + dense linear	RBF KRR residual

4.10 Rank nomination and independent gate

All 12 candidates are evaluated on the rank split. The lowest rank-set MSE becomes the **nominee**.

A simpler fallback is separately selected from

$$\mathcal{S} = \{\texttt{structured\_base}, \texttt{dense\_base}, \texttt{kernel\_poly}, \texttt{kernel\_rbf}\}$$

using rank-set performance.

If nominee and fallback differ, the gate split compares their per-sample errors. Let

$$e_i^{(f)} = \frac{1}{K}\|\hat{y}_i^{(f)} - y_i\|_2^2, \qquad e_i^{(n)} = \frac{1}{K}\|\hat{y}_i^{(n)} - y_i\|_2^2$$

and

$$d_i = e_i^{(f)} - e_i^{(n)}.$$

The nominee is accepted only when

$$\bar{d} - 1.96\frac{s_d}{\sqrt{n}} > 0$$

and its relative mean-error gain exceeds 5×10^{-4}. Otherwise the simpler fallback is retained.

This is a practical paired confirmation rule. It is not presented as a new multiple-testing theorem; its purpose is to prevent a rank-set winner from automatically becoming the deployed model.

4.11 Independent empirical release screen

After routing is complete, the chosen model is evaluated on the release-screen split. For each release-screen sample,

$$\ell_i = \frac{\|\hat{y}_i - y_i\|_2^2}{\mathbb{E}_{\text{screen}}\|y\|_2^2 + \epsilon}.$$

MX5 draws 600 nonparametric bootstrap resamples of the $n = 256$ sample losses and calculates the mean loss for each resample. Let $U_{0.95}$ denote the 95th percentile of those bootstrap means. The transformation is released only if

$$U_{0.95} \le \tau, \qquad \tau = 10^{-3}.$$

The software retains legacy names such as `certification`, `cert_nmse`, `cert_upper95`, and the prediction flag `require_certified=True`. In this manuscript these are described as **empirical release-screen** quantities to avoid implying a formal statistical certificate. When `require_certified=True`, the implementation refuses prediction if the empirical release rule is not met.

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5. Randomized Experimental Design

5.1 Random inputs

All scientific evidence in this study uses randomly generated data. No real-world dataset or fixed hand-curated signal corpus is used.

Inputs are sampled from three complex distributions:

- QPSK;
- normalized 16-QAM;
- circular complex Gaussian.

Depending on the transformation family, inputs also receive independently generated amplitude and phase nuisance jitter.

5.2 Transformation families

Eight random mechanism families form the frozen benchmark.

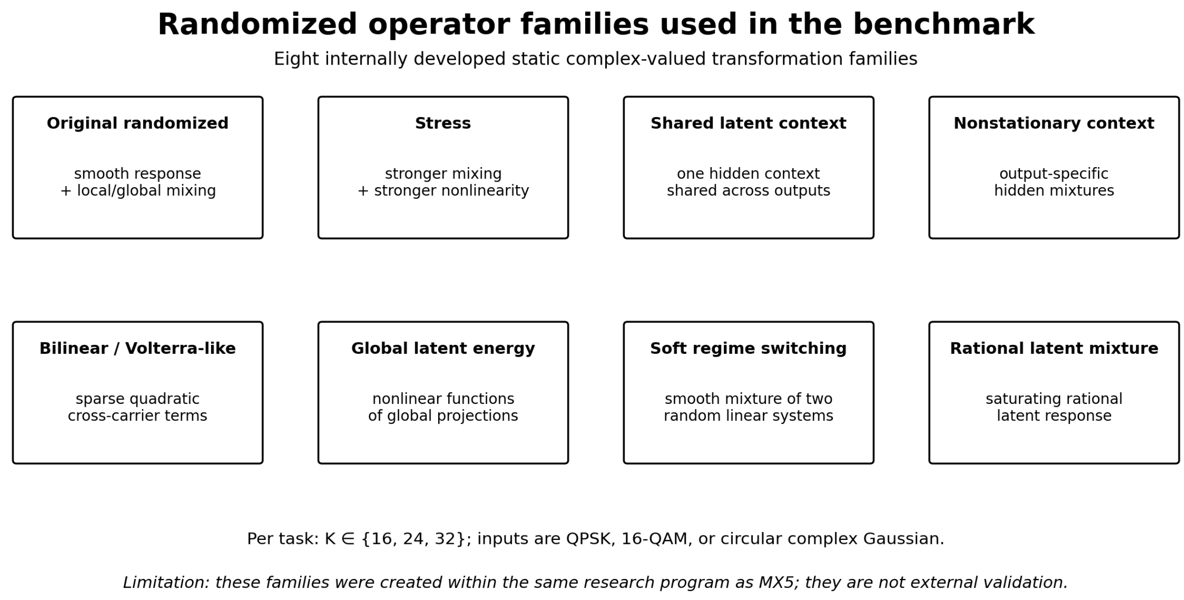

Figure 3: Eight internally developed randomized operator families used in the benchmark. These families provide heterogeneous static complex-valued mechanisms but do not constitute external validation.

Original randomized. Smooth gain/phase response, local circular mixing, low-rank long-range coupling, nonlinear even/odd carrier coupling, and one of several pointwise nonlinearities.

Stress. A stronger version of the original family with increased cross-carrier mixing, phase/gain variation, nonlinear coupling, and pointwise distortion.

Shared latent context. A common hidden circular mixture drives a nonlinear context applied across outputs.

Nonstationary per-output context. Different outputs depend on different random hidden carrier combinations.

Bilinear / Volterra-like. Sparse random quadratic interactions of the form

$$z_a \overline{z_b}$$

are injected into randomly chosen output carriers.

Global latent energy. Several unknown complex projections are transformed through nonlinear energy responses and remixed into the output.

Soft regime switching. Two different random linear systems are blended using a smooth sample-dependent gate.

Rational latent mixture. Unknown global projections are transformed through a saturating rational map

$$h(c) = \frac{c}{1 + 0.6|c|^2}$$

and linearly remixed into the output.

The router is not given the family label.

5.3 Development, Freeze, and Internal-Validation Boundary

The method was evolved through repeated random-data hypothesis tests. Developmental failures led to the introduction of harder random transformation families and to changes in the candidate pool. This adaptive development phase is reported in Section 7 but is **not counted as independent confirmation**.

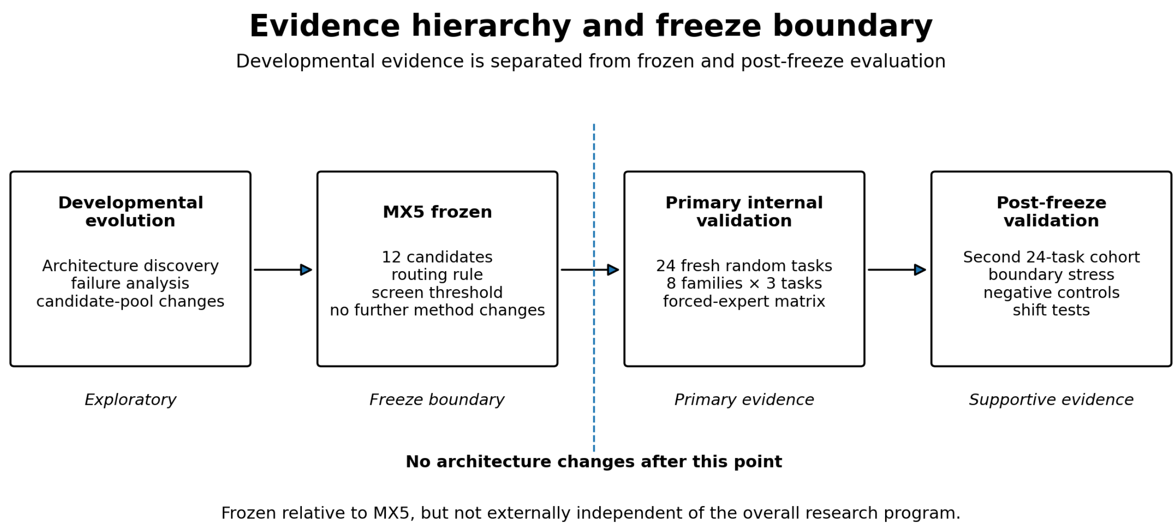

Figure 4: Evidence hierarchy and freeze boundary. Developmental evidence is separated from the primary frozen internal-validation cohort and post-freeze supporting analyses.

The frozen internal-validation boundary was defined when MX5, its twelve-candidate pool, the six data roles, the gate rule, the bootstrap count, and the release threshold $\tau = 10^{-3}$ were frozen. A new 24-task seed manifest was then generated. Those task labels were not used to alter MX5.

After the 24-task benchmark, additional challenge families were generated for stress testing. Because those tests occurred after the method freeze, they provide useful robustness evidence. They are nevertheless reported separately from the primary 24-task internal-validation estimands to avoid silently expanding the primary internal-validation dataset after seeing results.

5.4 Frozen Internal-Validation Benchmark

The eight primary transformation families were created within the same research program that produced MX5. Some families were introduced after developmental failures. The benchmark is therefore **not independent of the candidate-pool development process**. The frozen seed manifest prevents direct tuning to the final test draws, but it does not create external independence at the generator-family level.

The final benchmark contains three independent random transformations per family, for a total of 24 tasks. Carrier dimension varies among

$$K \in \{16, 24, 32\},$$

and the input distribution is independently chosen from the QPSK, 16-QAM, and complex Gaussian families according to each generator.

The final seed manifest was frozen before the reported evaluation.

5.5 Post-Freeze Structural-Boundary Stress Family

The primary random families often produce large margins between the best and second-best structural candidates. That makes them useful for evaluating routing accuracy but weak for testing whether independent gate evidence changes decisions near a structural boundary.

A separate post-freeze challenge family was therefore defined **without modifying MX5**. Each task combines two mechanisms that favor different experts:

1. sparse bilinear interactions, which favor polynomial-kernel structure; and
2. rational latent-mixture interactions, which favor latent-projection structure.

For each random task, the mechanism mixture weight is sampled from 0.35–0.65 and overall nonlinear strength from 0.20–0.42. Input type is randomly chosen from QPSK, 16-QAM, and complex Gaussian distributions, while $K \in \{16, 24, 32\}$. Twelve seeds were generated before inspecting the task outcomes. The same train/tune/rank/gate/release-screen/test sizes used by the primary benchmark were retained.

This challenge is **not external validation**: it was authored inside the same research program after the criticism motivating the ablation. Its purpose is narrower—to stress structural near-ties and compare full MX5, rank-only routing, and same-split release-screen reuse without changing the frozen model.

5.6 External baselines

Three fixed baselines are reported on all 24 test tasks:

1. **Dense complex linear:** symbolic mask followed by dense complex ridge regression.
2. **RBF-KRR:** dense-linear prediction plus an RBF-kernel residual model.
3. **Polynomial-KRR:** dense-linear prediction plus a degree-2/3 polynomial-kernel residual model.

The two kernel models are also members of the MX5 candidate pool. This is intentional: the comparison measures the value of adaptive structural routing relative to forcing one model family globally.

5.7 Statistical Analysis

Because the 24 tasks are nested within eight transformation families, the manuscript treats **family (n = 8) as the conservative inferential unit** for across-family claims. Task-level results remain useful descriptively, but family-level Wilcoxon tests and family-blocked bootstrap summaries are given greater interpretive weight. With only eight clusters, exact nonparametric inference is coarse; the paper therefore emphasizes effect sizes and consistency over thresholded p-values.

The benchmark has a hierarchical structure: individual input-output samples are nested within random tasks, and tasks are nested within **eight transformation families**. Treating all 24 tasks as fully exchangeable can understate uncertainty when within-family tasks are more similar than between-family tasks. This revision therefore treats **transformation family (n = 8) as the primary unit of inference** for across-method comparisons.

For each family, task-level NMSE values are first averaged within family. H1 is then evaluated by a paired one-sided exact Wilcoxon signed-rank test across the eight family-level means. Because three prespecified fixed baselines are tested, Holm correction is applied [13].

Original-budget baseline	Family-level one-sided p	Holm-adjusted p
Dense complex linear	0.003906	**0.011719**
RBF-KRR	0.003906	**0.011719**
Polynomial-KRR	0.031250	**0.031250**

Thus the original-budget H1 comparisons remain significant at 0.05 even under the more conservative family-level analysis. The previously reported 24-task Wilcoxon tests are retained only as sensitivity analyses and should not be interpreted as the primary inferential result.

A family-blocked bootstrap, resampling the eight transformation families rather than the 24 tasks independently, gives a 95% interval of approximately 6.3×10^{-5} **to** 5.65×10^{-4} for the primary overall mean MX5 NMSE. The analogous family-blocked bootstrap interval for release coverage is approximately **79.2%–100%**. These cluster-aware intervals are preferred to task-IID intervals when discussing uncertainty across the generator families.

The task-level exact Clopper-Pearson interval for 22/24 releases (73.0%–99.0%) is retained only as an IID descriptive calculation. It does not account for family clustering. Similarly, the simple zero-event upper bounds for 0/22 or 0/44 release violations assume independent Bernoulli trials and are not cluster-valid guarantees. If family is treated as the independent unit, only eight family clusters are available, making rare-failure inference necessarily weak.

The empirical bootstrap release screen is not a formal confidence bound. It is a fixed operational statistic computed after routing is frozen. Formal selective conformal or e-value procedures would be required for distribution-free finite-sample guarantees under their stated assumptions [7,8,14].

6. Results

6.1 Primary Frozen Internal-Validation Results

The family table reports **both all-task and released-only quantities**. Thus a family maximum can exceed the 10^{-3} release target when the corresponding task was rejected; released-only columns show the operating behavior after screening.

Family	Tasks	Released	All-task mean NMSE	All-task max NMSE	Released mean NMSE	Released max NMSE	Selected expert(s)
Original randomized	3	3	1.269e-05	2.340e-05	1.269e-05	2.340e-05	dense_base (2), structured_base (1)
Stress	3	3	3.520e-06	6.802e-06	3.520e-06	6.802e-06	structured_base (2), dense_base (1)
Shared latent context	3	3	2.276e-04	2.444e-04	2.276e-04	2.444e-04	dense_shared (3)
Nonstationary per-output context	3	2	6.081e-04	1.032e-03	3.961e-04	4.125e-04	kernel_poly (3)
Bilinear / Volterra-like	3	3	3.873e-06	5.729e-06	3.873e-06	5.729e-06	kernel_poly (3)
Global latent energy	3	3	2.633e-04	5.292e-04	2.633e-04	5.292e-04	dense_latent (3)
Soft regime switching	3	3	4.816e-06	5.906e-06	4.816e-06	5.906e-06	kernel_poly (3)
Rational latent mixture	3	2	1.124e-03	2.678e-03	3.471e-04	6.397e-04	dense_latent (2), masked_latent (1)

Across all 24 tasks, including predictions from the two tasks that were ultimately rejected,

$$\overline{\mathrm{NMSE}} = 2.810e - 04,$$

with a **family-blocked** 95% bootstrap interval of approximately

$$[6.31 \times 10^{-5},\ 5.66 \times 10^{-4}].$$

This interval resamples the eight transformation families and retains all three tasks within each sampled family. It is preferred to the narrower task-IID bootstrap interval reported in earlier drafts.

The system released 22/24 tasks:

$$\text{coverage} = 91.67\%.$$

Among those 22 released transformations:

- mean test NMSE was **1.379e-04**;
- family-blocked 95% bootstrap interval for mean released NMSE was approximately **[3.86e-05, 2.49e-04]**;
- median test NMSE was **1.579e-05**;
- worst released test NMSE was **6.397e-04**.

The worst released value remained below the predeclared 10^{-3} release threshold.

The two rejected tasks subsequently tested at approximately 1.03×10^{-3} and 2.68×10^{-3}. Thus, in this frozen benchmark, abstention prevented both above-threshold cases from being represented as supported predictions.

Because release outcomes are nested within eight transformation families, the preferred uncertainty summary is the family-blocked bootstrap interval of approximately **79.2%–100%** reported in Section 5.7. The exact 22/24 Clopper-Pearson interval of 73.0%–99.0% is retained only as an IID descriptive calculation and should not be interpreted as cluster-valid uncertainty.

Likewise, zero observed violations among 22 releases is not equivalent to zero deployment risk. A simple one-sided exact 95% upper bound for a Bernoulli violation probability after 0/22 events is approximately **12.7%**. The value is reported to prevent overinterpretation of a small zero-event sample; MX5's bootstrap screen is an empirical filter, not a formal distribution-free certificate.

6.2 Router behavior reflects mechanism structure

MX5 selected six different candidate types in the frozen benchmark:

Selected expert	Number of tasks
`kernel_poly`	9
`dense_latent`	5
`structured_base`	3
`dense_base`	3
`dense_shared`	3
`masked_latent`	1

The pattern is qualitatively meaningful despite the family label never being passed to the router. All three bilinear tasks selected polynomial KRR. All three regime-switching and all three nonstationary tasks also selected polynomial KRR in this particular frozen sample. All three global-energy tasks selected `dense_latent`, all three shared-context tasks selected `dense_shared`, and the rational family used latent models at both placements.

This result supports the central motivation of structural routing: different random mechanisms favor materially different inductive biases.

The table above alone does not distinguish the benefit of **routing** from the benefit of maintaining a larger candidate pool. Section 6.5 therefore evaluates every frozen candidate as a forced model on every untouched primary test task. This post-hoc diagnostic was not used to alter MX5.

6.3 Prespecified Baseline Comparison with Family-Level Inference

Raw MX5 predictions are compared with the three prespecified fixed baselines on all 24 primary tasks, including the two tasks that the selective operating rule would reject.

Method	Mean task NMSE	Mean-error ratio to MX5	Task-wise MX5 wins
Dense complex linear	4.418e-03	15.72×	24/24
RBF kernel ridge	3.358e-03	11.95×	24/24
Polynomial kernel ridge	3.192e-03	11.36×	15/24
STABLE-Wave MX5	**2.810e-04**	**1.00×**	—

For inference, however, the eight transformation families are the analysis units. The family-level paired one-sided Wilcoxon tests yield Holm-adjusted p-values of **0.0117**, **0.0117**, and **0.03125** for dense linear, RBF-KRR, and polynomial-KRR, respectively. H1 therefore remains supported within the original-budget design after accounting for family clustering.

The much smaller task-level p-values reported in earlier manuscript versions are no longer used as the primary evidence because they treat the three tasks within a family as exchangeable independent units.

The polynomial baseline deserves special interpretation. MX5 does not beat polynomial KRR because polynomial KRR is excluded; it frequently **uses polynomial KRR internally**. The benefit is the ability to retain it when appropriate and switch to another structural hypothesis when a different representation generalizes better.

6.4 Equal-Total-Observation-Budget Diagnostic

The original fixed baselines use their training/tuning data for parameter estimation, whereas MX5 spends additional non-test observations on ranking, gating, and release screening. To test whether the headline advantage is merely a larger observation budget, each fixed baseline was refit post hoc using **all 1,152 non-test observations per task** (train + tune + rank + gate + release-screen samples), while the untouched test sets remained unchanged.

Equal-budget method	Mean test NMSE	Mean-error ratio to MX5	Task-wise MX5 wins
Dense complex linear	4.295e-03	15.28×	22/24
RBF-KRR	2.869e-03	10.21×	16/24
Polynomial-KRR	2.784e-03	9.91×	15/24
STABLE-Wave MX5	**2.810e-04**	**1.00×**	—

The mean differences remain large, but the family-level exact Wilcoxon tests are much weaker:

Equal-budget baseline	Family-level one-sided p	Holm-adjusted p
Dense complex linear	0.019531	**0.058594**
RBF-KRR	0.054688	**0.109375**
Polynomial-KRR	0.054688	**0.109375**

Accordingly, the equal-budget diagnostic does **not** support a conventional 0.05-level claim that MX5 is superior to these pooled-data fixed baselines after family clustering and multiplicity correction. The correct interpretation is narrower: MX5 has substantially lower mean error on this benchmark, but part of the original inferential advantage reflects the procedural use of extra observations for structural adaptation rather than a clean same-budget model-versus-model comparison.

This post-hoc analysis is deliberately reported even though it weakens the headline result, because it separates **structural routing performance** from **data-budget advantage**.

6.5 Task-Level Effect Sizes and Outlier Robustness

Mean error ratios can be dominated by a small number of tasks on which one baseline fails severely. We therefore also report task-wise baseline/MX5 error ratios.

Baseline	Median task-wise error ratio	Geometric-mean error ratio	Smallest ratio	Largest ratio
Dense complex linear	2.17×	11.74×	1.00×	353.42×
RBF kernel ridge	2.16×	7.68×	1.02×	262.99×
Polynomial kernel ridge	1.87×	6.43×	1.00×	248.90×

All ratios are baseline NMSE divided by MX5 NMSE, so values above one favor MX5. The median improvements of approximately 1.87×–2.17× show that the aggregate advantage is not solely an artifact of a few catastrophic baseline failures. The larger geometric-mean ratios reflect genuine heavy-tailed baseline failures on some mechanism families.

6.6 Full Forced-Expert Matrix and Structural-Selection Regret

To address whether MX5 benefits merely from having a large expert pool, the frozen procedure was rerun on the same primary seed manifest solely so that predictions from **all twelve already-frozen candidates** could be retained and scored on the untouched test sets. No test result was used to alter model fitting, routing, gating, or release decisions. For each task, the retrospective *test oracle* is defined only for analysis as the candidate with the smallest test NMSE.

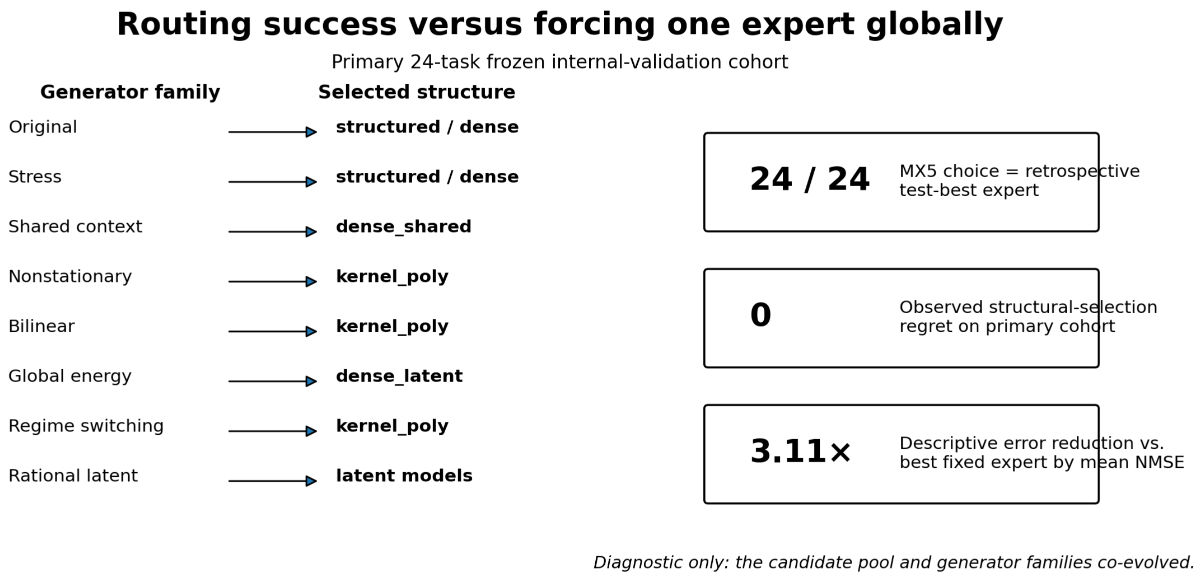

Figure 5: Routing versus forcing one expert globally. Different transformation mechanisms favor different structures; the primary forced-expert diagnostic measures structural-selection regret directly.

The expert selected by MX5 using train/tune/rank/gate/release-screen data matched the retrospective test-best candidate on

$$\boxed{24/24}$$

tasks. The observed mean structural-selection regret,

$$\frac{1}{24}\sum_t \left(L_{\mathrm{MX5},t} - \min_j L_{j,t}\right),$$

was **0.000e+00**. In other words, no primary task showed positive test regret relative to the twelve-candidate oracle. This is a strong empirical routing result, but it is not a theorem that MX5 will recover the oracle on future tasks.

Forced candidate	Mean test NMSE	Median test NMSE	Test-best tasks	MX5 strict wins	Ties with MX5
`dense_latent`	8.726e-04	4.789e-04	5/24	19/24	5/24
`masked_latent`	2.814e-03	1.543e-03	1/24	23/24	1/24
`kernel_poly`	3.192e-03	3.531e-04	9/24	15/24	9/24
`kernel_rbf`	3.358e-03	3.705e-04	0/24	24/24	0/24
`dense_base`	3.663e-03	1.810e-04	3/24	21/24	3/24
`dense_shared`	3.689e-03	2.192e-04	3/24	21/24	3/24
`structured_per_output`	3.772e-03	2.059e-04	0/24	24/24	0/24
`dense_per_output`	3.800e-03	2.642e-04	0/24	24/24	0/24
`structured_base`	3.959e-03	2.095e-04	3/24	21/24	3/24
`dense_linear`	4.418e-03	6.391e-04	0/24	24/24	0/24
`structured_shared`	5.557e-03	4.577e-04	0/24	24/24	0/24
`structured_linear`	1.097e-02	8.505e-03	0/24	24/24	0/24

The retrospectively best *single fixed* candidate by mean test NMSE was `dense_latent` at **8.726e-04**, whereas adaptive MX5 achieved **2.810e-04**, a descriptive **3.11×** error reduction. Because the identity of this best fixed candidate is chosen after inspecting the 24 test tasks, that particular 3.11× comparison is treated as descriptive rather than as a prespecified inferential test.

Earlier task-level post-hoc Wilcoxon comparisons against all twelve forced candidates are not used as inferential evidence in this revision because they inherit the same family-clustering problem. The 24-by-12 matrix and 24/24 oracle-match result are therefore treated as **descriptive structural-routing diagnostics**. These analyses strengthen the interpretation that routing contributes beyond the mere presence of a rich expert pool, but they are explicitly secondary to the three prespecified external-baseline comparisons.

The complete 24-by-12 forced-expert matrix is supplied as a machine-readable supplement.

6.7 Data-Role Separation Ablation: Full MX5 Versus Naive Reuse

The central procedural motivation for MX5 is to avoid repeatedly reusing the same held-out observations. The previous manuscript described this as a safeguard but did not directly test whether the six-way separation changes empirical outcomes. We therefore performed a post-freeze ablation on the same 24 primary tasks.

Three procedures were compared without changing the fitted candidate pool:

1. **Full MX5:** rank nomination, independent gate, independent release-screen split.
2. **Rank-only + independent screen:** accept the rank winner directly, omit the independent gate, but retain a separate release-screen split.
3. **Naive reused validation:** accept the rank winner directly and reuse the **rank split itself** to estimate the release-screen statistic.

The result is a useful negative finding:

Diagnostic	Full MX5 vs rank-only	Full MX5 vs reused-validation
Same selected expert	24/24	24/24
Same release decision	24/24	24/24
Released tasks	22/24 vs 22/24	22/24 vs 22/24
Released-task threshold violations	0/22 vs 0/22	0/22 vs 0/22
Test prediction difference	None	None

The gate never overturned the rank nominee on these 24 tasks, and reusing the rank split for release screening did not change any binary release decision.

This result **does not invalidate data-role separation** as a design principle. It shows something more specific: on the final benchmark, candidate margins were large enough that the extra separation did not alter outcomes. Therefore the paper cannot claim that six-way splitting is empirically superior to a simpler reuse design from these data alone.

The six-way design should instead be interpreted as a **conservative anti-reuse protocol** whose incremental benefit remains unproven on the present benchmark. Demonstrating that benefit would require a prospectively defined benchmark concentrated near structural and release boundaries, or a formal analysis of selection-induced optimism.

6.8 Post-Freeze Structural-Boundary Stress Ablation

The ordinary 24-task ablation produced no difference at all between full MX5 and simpler reuse procedures. To test whether the gate matters when structural margins are deliberately smaller, the frozen method was evaluated on the 12-task boundary family described in Section 5.5.

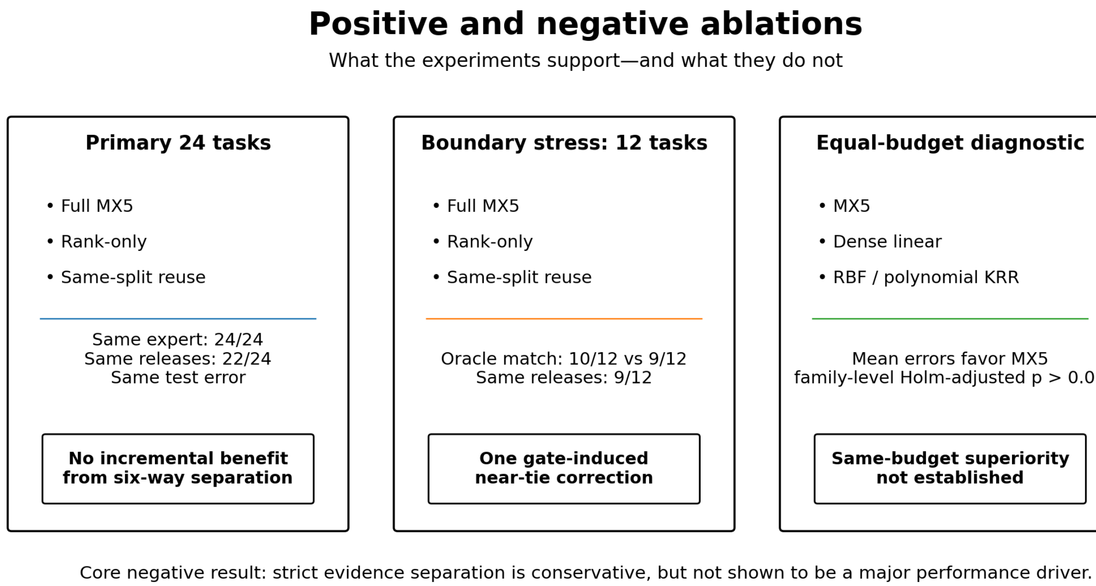

Figure 6: Positive and negative ablations. Routing succeeds internally, while six-way separation and equal-budget superiority are not established as major performance advantages.

Quantity	Full MX5	Rank-only + independent screen	Rank winner + reused rank screen
Released tasks	9/12	9/12	9/12
Released tasks above 10^{-3} on test	0/9	0/9	0/9
Mean raw test NMSE	8.263e-04	8.264e-04	8.264e-04
Structural choices matching retrospective test oracle	10/12	9/12	9/12
Mean structural-selection regret	4.416e-06	4.508e-06	4.508e-06

The independent gate changes the rank-only structural choice on **1/12** task. In that task, the rank split nominates `dense_linear`, whereas full MX5 ultimately selects `kernel_poly`; `kernel_poly` is also the retrospective test-best candidate. The test NMSE difference is small (7.906×10^{-4} versus 7.917×10^{-4}), so this is evidence that the gate can resolve a near-tie, not evidence of a large practical gain.

Release behavior is unchanged: all three procedures release the same **9/12** tasks, and none of the released tasks exceeds 10^{-3} on test. Reusing the rank split changes the numerical release-screen statistic—reuse minus independent-screen upper values range from **-6.48e-05 to 1.89e-04**, with mean **2.53e-05**—but those shifts do not cross a release boundary in this 12-task sample.

The boundary family therefore refines, rather than reverses, the primary negative result:

> **Independent gate evidence can change a near-tie structural decision, but the current simulations still do not demonstrate that six-way evidence separation reduces false releases or materially improves aggregate error.**

A stronger empirical test of anti-reuse benefits would require many more prospectively frozen tasks concentrated around rank and release boundaries, preferably from independently authored generators.

6.9 Release-Screen/Test Agreement

Across the 24 primary frozen tasks, the descriptive Pearson correlation between release-screen NMSE and untouched test NMSE was

$$r = 0.99987.$$

This high value is expected to some extent because the release-screen and test sets are independent draws from the **same frozen transformation and input distribution**. It is therefore evidence of within-task replication, not evidence of robustness to distribution shift and not evidence of formal calibration.

6.10 Expanded Negative Controls

The original artifact contained four independently permuted controls. To strengthen this sanity check after MX5 was frozen, eight additional controls were generated—one from each transformation family. In every control, the input-output pairing was independently permuted in **each label-bearing pre-test split** before fitting, while the final test pairing remained unpermuted.

The result is now:

$$\boxed{12/12 \text{ negative controls rejected}}$$

The eight new release-screen upper values ranged from approximately 1.031 to 1.048 NMSE, and test NMSE against the true pairing remained approximately 1.009–1.032. In all eight new runs the router collapsed to the simple `structured_linear` model and the release screen refused deployment.

These controls show that MX5 does not obtain its reported low error from output marginals, routing mechanics, or the candidate pool alone when the task relationship is destroyed. They remain a sanity check rather than a formal hypothesis test about every possible null mechanism.

6.11 Secondary Post-Freeze Validation Cohort

A second validation cohort was generated **after MX5 was frozen and without any model, candidate-pool, threshold, gate, or generator-family changes**. It contains 24 new random tasks: three per family, with one task at each of $K = 16, 24, 32$. This cohort was not used to modify MX5 and is reported separately from the primary internal-validation set.

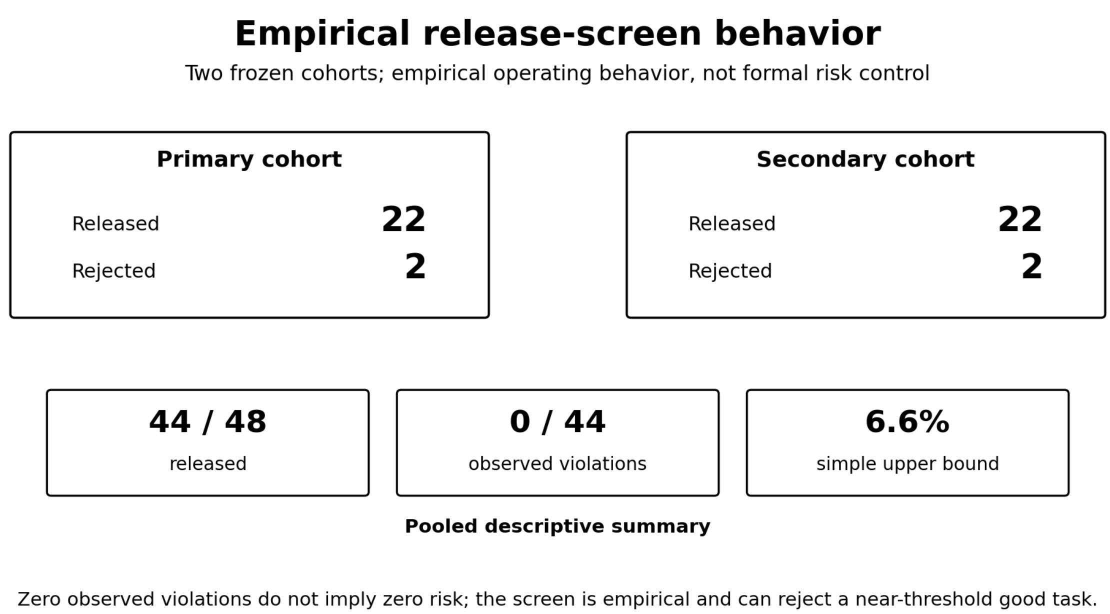

Figure 7: Empirical release-screen behavior across the primary and secondary frozen cohorts. Zero observed released-task violations are reported as empirical evidence, not as a formal risk guarantee.

Family	Tasks	Released	All-task mean NMSE	Released mean NMSE	Released max NMSE	Selected expert(s)
Original randomized	3	3	2.165e-05	2.165e-05	4.156e-05	structured_base (2), dense_base (1)
Stress	3	3	2.417e-05	2.417e-05	6.450e-05	structured_base (2), dense_base (1)
Shared latent context	3	3	5.233e-04	5.233e-04	7.145e-04	dense_shared (3)
Nonstationary per-output context	3	3	4.372e-04	4.372e-04	5.475e-04	kernel_poly (3)
Bilinear / Volterra-like	3	3	3.504e-06	3.504e-06	7.463e-06	kernel_poly (3)
Global latent energy	3	3	3.201e-04	3.201e-04	4.760e-04	dense_latent (3)
Soft regime switching	3	3	9.295e-06	9.295e-06	1.602e-05	kernel_poly (2), dense_base (1)
Rational latent mixture	3	1	1.564e-03	6.783e-04	6.783e-04	masked_latent (3)

Overall, the second cohort produced:

- **22/24 releases (91.67%)**;
- **0/22 released-task threshold violations**;
- raw mean NMSE **3.629e-04**;
- mean released NMSE **2.135e-04**;
- worst released NMSE **7.145e-04**.

The two rejected tasks were both rational-mixture cases. One was a genuine above-target failure with test NMSE 3.102×10^{-3}. The other was a conservative false rejection: its release-screen upper value was 1.061×10^{-3}, while its final test NMSE was 9.108×10^{-4}. This is an important counterexample to any suggestion that the screen is perfectly calibrated: it can sacrifice coverage near the decision boundary.

MX5 again beat dense linear and RBF-KRR on **24/24** tasks and polynomial-KRR on **16/24**. Paired one-sided Wilcoxon p-values were 9.11e-06, 9.11e-06, and 5.85e-05, respectively.

If the two frozen cohorts are pooled **descriptively**, they contain 48 tasks with 44 releases. The resulting release-rate point estimate is 91.67% with exact 95% interval **80.0%–97.7%**. No released task in either cohort exceeded 10^{-3}, so 0/44 violations were observed. A simple one-sided zero-event 95% upper bound is still **6.6%**, which remains far too large to support a high-stakes safety claim. The pooled calculation is therefore a precision summary, not a formal deployment guarantee.

6.12 Post-Freeze Composed Transformations

After MX5 was frozen, a new random generator was constructed by composing several mechanisms rather than merely increasing parameter strength within an existing family. It combined linear mixing, rational latent response, latent-energy response, bilinear effects, and soft regime behavior.

MX5 released all three post-freeze tasks. Untouched test NMSE values were

$$4.76 \times 10^{-6}, \quad 2.82 \times 10^{-6}, \quad 5.52 \times 10^{-6}.$$

All three selected `dense_latent`.

These results provide evidence against the narrowest form of generator-specific overfitting, although three tasks are not sufficient to establish universal compositional generalization.

6.13 Input-Distribution Shift

Three additional post-freeze tests changed the input distribution between model development/release screening and final evaluation.

- global-energy: Gaussian → QPSK;
- rational: Gaussian → 16-QAM;
- bilinear: Gaussian → QPSK.

All three remained accurate. The largest shifted test NMSE was 1.90×10^{-4}, on the global-energy task.

This experiment is useful but limited. The release screen itself was computed in-distribution; therefore the result demonstrates empirical robustness to these shifts, not a formal guarantee under arbitrary distribution shift.

6.14 Sample Efficiency

Two fresh tasks were evaluated at increasing training-set sizes.

Family	Train pairs	Selected expert	Released?	Release-screen upper value	Test NMSE
Original	128	`structured_per_output`	Yes	4.43e-5	4.52e-5
Original	256	`dense_base`	Yes	3.03e-6	2.88e-6
Original	512	`dense_base`	Yes	1.61e-6	1.56e-6
Global energy	128	`kernel_poly`	No	1.37e-2	1.32e-2
Global energy	256	`kernel_poly`	No	1.10e-2	1.06e-2
Global energy	512	`dense_latent`	Yes	2.13e-4	2.07e-4

The global-energy case illustrates a desirable selective behavior. At 128 and 256 training pairs, the evidence does not support the latent structure strongly enough and the system rejects the task. At 512 pairs, the router changes structural hypothesis, passes the release screen, and test error drops by roughly two orders of magnitude.

6.15 Computational Cost

The reference implementation required a mean of **9.44 s** and median of **9.87 s** per final benchmark task in the recorded development environment. Hardware was not standardized as part of the frozen artifact, so these numbers should be interpreted as reproducibility diagnostics rather than a platform-independent speed benchmark.

6.16 Evidence Summary

The evidence supports a narrower set of conclusions than an algorithm-centric reading would suggest.

Question	Evidence	Conclusion
Can heterogeneous structural models be routed accurately on the internal generator suite?	Primary forced-expert matrix: MX5 pre-test choice equals retrospective test-best expert on 24/24 tasks	**Yes, internally and descriptively**
Does MX5 outperform prespecified fixed baselines under the original information budget?	Large mean-error reductions; family-level tests remain significant after Holm correction	**Supported within the internal benchmark**
Does MX5 retain statistically significant superiority when fixed baselines receive equal total observations?	Means favor MX5, but family-level Holm-adjusted tests exceed 0.05	**Not established**
Does six-way separation materially improve routing or release behavior?	Primary 24 tasks: no difference. Boundary stress: one structural correction in 12 tasks, no release-decision difference	**Not demonstrated as a material advantage**
Does the empirical release screen behave sensibly?	0/44 released-task threshold violations across two frozen cohorts; one conservative false rejection; zero-event upper bound about 6.6%	**Promising empirical behavior, not formal risk control**

Question	Evidence	Conclusion
Does the method fabricate signal when pairings are destroyed?	12/12 permuted controls rejected	**Strong sanity-check result**
Is external generalization established?	No independent generator suite, public benchmark, real measured system, or temporal dynamical benchmark	**No**

The strongest positive result is therefore **accurate internal structural routing**. The strongest negative result is that the present evidence does **not** show that the elaborate six-way split is necessary or that adaptive routing is superior under equal information budgets.

7. Evolutionary Development and Ablative Evidence

MX5 was not designed as a single monolithic architecture. It emerged from repeated randomized hypothesis-test cycles. A component was retained only when it improved held-out random performance or repaired a reproducible failure mode.

The main accepted changes were:

Evolution	Observed role
Random hard-family expansion	Revealed failures hidden by the original four-family benchmark
Symbolic + complex linear hierarchy	Removed smooth/global linear structure before nonlinear fitting
Structured banded + low-rank bridge	Efficiently modeled local and low-rank cross-carrier coupling
Topology-adaptive residual discovery	Reduced error when relevant carrier separations varied by task
Shared/per-output context experts	Addressed common vs. output-specific latent context
Latent projection expert	Reduced global-mixture failures from roughly 0.008–0.030 NMSE to the 10^{-4} scale in successful cases
Dual latent placement	Allowed nonlinear structure to be modeled before or after dense linear correction
Polynomial/RBF experts inside router	Allowed classical nonlinear models to win when appropriate
Independent release-screen split	Converted the system from forced prediction to selective release
Post-freeze controls	Tested permutation failure, new compositions, sample efficiency, and input shift

Several plausible extensions were rejected:

- a generic two-layer MLP residual fallback, which overfit relative to structured experts;
- a direct rational latent network, which became unstable at larger K;
- a variable-projection rational model, which was more stable but not sufficiently accurate;
- a random nonlinear feature bank, which did not repair the hard latent failures;
- a spectral/cross-covariance latent discovery method, which was fast but inaccurate;
- increasing the number of training pairs without changing representation, which did not fix structural mismatch.

This developmental record matters because it shows that final complexity was not added indiscriminately. Neural components from earlier SAIR-Wave variants were removed when semi-parametric structural models generalized better. Conversely, a classical polynomial kernel expert was added when a deliberate bilinear challenge demonstrated that it was the correct inductive bias.

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8. Discussion

The results are best interpreted as a study of **routing and evidence separation under controlled synthetic heterogeneity**, not as validation of a new general-purpose operator-identification procedure. The method works very well inside the generator program, but the negative ablations materially constrain the stronger procedural story.

8.1 Structural routing rather than universal approximation

The frozen results suggest that the principal advantage of MX5 is not one unusually powerful predictor. It is the refusal to assume that one representation is universally appropriate.

The expert selections align with the mechanisms that generated the data:

- polynomial kernels dominate sparse bilinear transformations;
- latent projection models dominate global-energy and rational mixture transformations;
- shared-context models dominate shared hidden circular mixtures;
- structured or dense topology models dominate the original and stress families.

Because the router does not observe family identity, those choices must be inferred from performance on independently sampled inputs from the same transformation.

This is the central empirical finding of the paper.

8.2 Routing Gain, Data Budget, and Candidate-Pool Capacity

The forced-expert matrix addresses one concern cleanly: the benefit is not explained by a single hidden champion in the twelve-model pool. `dense_latent` is the strongest fixed candidate by retrospective mean test NMSE, yet forcing it globally is still substantially worse than adaptive routing, and MX5's pre-test choice matches the retrospective test-best candidate on all 24 primary tasks.

A different concern remains: **MX5 spends more observations on structural decision-making**. The equal-total-observation-budget diagnostic gives the fixed baselines all 1,152 available non-test observations. Their mean errors remain substantially above MX5 overall, but after reducing inference to eight family clusters and applying Holm correction, the differences are no longer conventionally significant.

Accordingly, the fairest interpretation is not "MX5 is definitively better than every fixed model under equal information." It is:

> **Within this synthetic setting, allocating observations to structural adaptation produces low error and excellent task-specific expert choice, but the present sample of eight families is insufficient to establish a same-total-budget statistical advantage over every strong fixed baseline.**

Future comparisons should prospectively equalize total observation budgets and evaluate cross-fitted routers that use data more efficiently.

8.3 What the Six-Way Split Does—and Does Not Yet Prove

The primary benchmark does **not** show an incremental benefit from the extra evidence splits. The rank winner is also the final MX5 choice on all 24 primary tasks; the independent gate never changes the winner; and reusing the rank sample for release screening changes no release decision.

The post-freeze structural-boundary family gives a slightly different result. Across 12 deliberately more ambiguous random tasks, the gate changes **1/12** structural decision and increases retrospective oracle matching from **9/12 to 10/12**. That single changed decision is directionally correct on untouched test data, but its error reduction is small. More importantly, full MX5, rank-only routing, and same-split reuse still release exactly **9/12** tasks and show the same released-task violation count.

The correct interpretation is therefore conservative:

> **Separate rank, gate, and release-screen roles reduce direct evidence reuse by design and can alter a near-tie structural decision, but the present experiments do not establish a material performance or release-safety advantage from six hard splits.**

This distinction matters because the procedural idea would be overstated if the paper claimed that six-way splitting was experimentally necessary. A decisive test would require a prospectively frozen boundary benchmark with enough independent generator families to estimate selection optimism and release errors directly.

8.4 Empirical Release Screening as an Operating Rule

MX5's selective operating contract is simple:

$$\text{release only if empirical release screen} \leq 10^{-3}.$$

In the primary frozen evaluation, 22 tasks passed this screen and all 22 subsequently remained below threshold on test. A second 24-task post-freeze cohort repeated the same 22/24 release count with 0/22 released-task threshold violations, while also showing one conservative false rejection near the threshold. The screen therefore behaved consistently in these cohorts but should not be described as perfectly calibrated.

That correspondence is encouraging, but the terminology must remain precise. The bootstrap percentile is an empirical screening statistic. It does not establish the finite-sample, distribution-free risk control supplied by modern conformal or e-value methods [7,8]. The present paper therefore reports an empirical release-screening experiment rather than a theorem.

8.5 Why the Release Screen Is Not Called a Certificate

Modern selective conformal and e-value methods can provide finite-sample risk guarantees under explicit assumptions [7,8,14]. MX5 does not implement those guarantees. Its screen bootstraps the mean normalized loss on an independent sample and compares an empirical upper percentile with a fixed engineering tolerance.

Retrofitting a conformal or e-value decision rule **after observing the MX5 results** and then presenting the old benchmark as validation of that new rule would be methodologically unsound. Such a change would define a new method. The present manuscript therefore keeps MX5 frozen, renames the mechanism *empirical release screening*, and makes formal risk control a separate prospective extension.

A future version could predeclare a bounded risk functional and replace the bootstrap rule with selective conformal risk control, an e-value procedure such as SCoRE, or a joint finite-sample selective certificate. That extension should then be evaluated on a new frozen seed manifest rather than retroactively on the current test results.

8.6 Relation to Broader System Identification

Recent work such as ASIA emphasizes that identifying a nonlinear system involves choices over representation, training, and validation rather than only fitting coefficients [10]. MX5 supports a complementary view:

> **model structure itself can be treated as a testable hypothesis, and the identification algorithm can allocate independent evidence to choosing, confirming, and accepting that hypothesis.**

This perspective is applicable beyond the exact candidate pool used here. A future implementation could replace the present experts with state-space, NARX, Gaussian-process, neural-operator, physics-informed, or application-specific grey-box models while preserving the same data-role architecture.

8.7 What the Results Establish—and What They Do Not

The strongest evidence concerns **procedure-level structural adaptation**. MX5 does not win because every internal expert is superior to classical alternatives. Indeed, polynomial KRR is selected on nine of the 24 final tasks. The advantage arises because the procedure can retain a classical model on mechanisms for which it is appropriate and choose a different structural hypothesis when another model generalizes better.

The study demonstrates that using six disjoint data roles is operationally feasible in a synthetic setting where new random observations are inexpensive. The direct ablation further shows that, on the present primary tasks, the extra gate and independent release-screen split do not change the selected expert or release outcome. It therefore does not establish that six-way splitting is empirically superior, necessary, or sample-optimal. In data-scarce applications, cross-fitting, reusable holdout methods, or formal selective-inference procedures may provide better efficiency.

Finally, the 24-task benchmark establishes performance on a *defined generator distribution*, not on “nonlinear systems in general.” This wording is intentional. The post-freeze composition and input-shift experiments widen the empirical support, but they remain simulation evidence.

8.8 Negative Results as a Scientific Contribution

Two null or weak results are central rather than incidental.

First, the primary benchmark provides **no empirical advantage from independent gate and release-screen roles** relative to simpler reuse procedures. Candidate margins are apparently large enough that all procedures make the same decisions. The boundary-stress experiment produces only one gate-induced correction in twelve tasks and no release-decision difference.

Second, once fixed baselines are allowed to consume the same total number of observations, the mean-error advantage of MX5 remains large but family-level inferential support is no longer conventionally significant after Holm correction.

These findings change the paper’s scientific interpretation. The evidence supports structural routing in heterogeneous synthetic tasks, but it does not support the stronger claim that the particular six-way protocol is necessary or that the adaptive procedure wins solely because of superior structural reasoning rather than because it spends additional observations on adaptation. Reporting these negative findings is part of the methodological contribution.

9. Limitations

First and most importantly, the benchmark remains internal to the research program. The second 24-task cohort improves precision, and the forced-expert matrix clarifies routing value, but the same eight generator families originate from the developmental program. The three post-freeze composed tasks are a useful held-out challenge, yet they were still authored within the same research program and are too few to function as an independent benchmark. These experiments therefore strengthen internal validity, not external validity. An independently authored random generator suite or measured dataset would materially strengthen the claim.

Second, all data are synthetic and randomly generated. This is an explicit design constraint and is useful for controlled mechanism testing, but it does not demonstrate performance on measured communication channels, biological systems, industrial processes, or other physical datasets.

Third, the evaluated operators are static across samples. They may be highly nonlinear across complex coordinates, but they do not contain temporal hidden state, feedback, or long memory. The broader nonlinear system-identification literature is relevant, yet the empirical object in this paper is a static input-output operator.

Fourth, the sample size remains modest for rare release-screen failures. The primary and secondary cohorts together contain 48 tasks and 44 releases. Observing 0/44 released-task violations reduces the simple one-sided zero-event 95% upper bound to approximately 6.6%, but that is still far too weak for a high-stakes safety claim.

Fifth, the empirical release screen is bootstrap-based and lacks a formal distribution-free finite-sample guarantee. Selective conformal or e-value methods provide relevant formal alternatives [7,8,14], but adopting one would define a new prospective method rather than retroactively strengthen MX5.

Sixth, the candidate pool and generator program co-evolved. Harder families were introduced after developmental failures. This is preferable to hiding those failures, but it means the expert pool was not designed independently of the simulation program. The three composed post-freeze tasks and three input-shift tasks are useful stress checks, but they remain too small to establish broad compositional or shift robustness.

Seventh, the full forced-expert matrix resolves one ambiguity but not external independence. It shows that routing adds value beyond forcing any one of the twelve candidates on the primary frozen tasks, yet the twelve-candidate pool itself remains a product of the developmental program.

Eighth, the 10^{-3} release tolerance is an engineering operating point. It was fixed before the primary frozen benchmark, but it is not a universally meaningful scientific constant. A future risk-coverage study should use a prespecified tolerance grid.

Ninth, the gate rule is heuristic. It uses a normal-approximation paired improvement criterion and a relative-gain floor. The separate gate sample reduces direct rank-set reuse, but the rule is not a theorem of post-selection validity.

Tenth, six hard data partitions are statistically expensive. This is feasible in synthetic simulation because fresh random observations are cheap. In scarce-data settings, cross-fitting, nested resampling, reusable holdouts, or formal post-selection methods may be more efficient.

Eleventh, MX5 contains a small nonconvex latent-projection optimizer. Although seeds are frozen, platform-level numerical variation is possible.

Twelfth, runtime was not benchmarked on standardized hardware. Kernel experts also scale poorly with sample count in the direct reference implementation because they solve dense kernel systems.

Thirteenth, the benchmark is hierarchically clustered. The primary and secondary cohorts contain three tasks per family, not 24 or 48 unrelated mechanisms. This revision moves family-level inference to the foreground, but only eight independent family clusters remain. That sharply limits the precision of statistical conclusions and rare-failure estimates.

Fourteenth, evidence for six-way separation remains weak. The naive-router ablation produces identical primary selections and releases. A post-freeze 12-task structural-boundary family shows one gate-induced structural correction (9/12 to 10/12 oracle matches), but no release-decision difference. The separation is therefore primarily a conservative methodological design, with only modest empirical evidence for incremental structural-selection benefit.

Fifteenth, equal-total-observation-budget evidence is weaker than the original-budget headline comparison. Although MX5 retains much lower mean NMSE, family-level multiplicity-corrected tests against equal-budget fixed baselines do not cross the conventional 0.05 threshold. Any claim of fixed-model superiority must preserve this distinction.

10. Reproducibility

The accompanying implementation contains:

- the frozen MX5 estimator;
- all random generator definitions;

- the final seed manifest;
- per-family result files;
- negative controls;
- post-freeze composed tests;
- distribution-shift tests;
- sample-efficiency tests;
- the evolutionary development log;
- source-lineage notes.

The principal software requirements are NumPy, SciPy, and PyTorch.

The packaged benchmark can be reproduced with:

```
pip install -r requirements.txt
python reproduce_mx5_final.py
```

A smaller fresh-random smoke test is available through:

```
python smoke_mx5.py
```

The final test split is not used by the MX5 fitting routine. Each benchmark task independently generates train, tune, rank, gate, release screening, and test samples from a frozen random transformation seed.

10.1 Reproducibility and Reporting Checklist

Item	Status
Frozen implementation supplied	Yes
Random generator code supplied	Yes
Final seed manifest supplied	Yes
Six data roles explicitly separated	Yes
Release tolerance fixed in code	Yes, 10^{-3}
Test labels excluded from fitting/routing/release screening	Yes
Negative controls supplied	Yes; original 4 plus 8 post-freeze family-wide controls
Secondary 24-task post-freeze cohort supplied	Yes
Full 24×12 forced-expert matrix supplied	Yes
Family-level clustered reanalysis supplied	Yes
Naive reused-validation router ablation supplied	Yes
12-task structural-boundary stress ablation supplied	Yes
Equal-total-observation-budget baseline diagnostic supplied	Yes
Post-freeze challenge results supplied	Yes
Developmental rejected branches documented	Yes
Hardware standardized for runtime claims	No
Real-world measured dataset	No, by study design
Formal conformal/e-value release guarantee	No; explicitly out of current MX5 claim
External prospective preregistration	No
Independently authored external generator suite	No

This checklist is included to make the evidentiary boundary auditable rather than implicit.

10.2 Data and Code Availability

The research artifact contains the frozen estimator, benchmark generators, final seeds, control experiments, result JSON files, and reproduction scripts. This revision additionally reports a machine-readable 24×12 forced-expert matrix, a 24-task secondary validation table, and eight new family-wide negative-control records. Because the paper uses synthetic random data, the dataset is generated from code rather than distributed as a fixed corpus. Reproducing the final benchmark therefore requires preserving both the generator implementation and the frozen seed manifest.

11. Conclusion

This study evaluated structural-hypothesis routing in a controlled program of synthetic static complex-valued nonlinear operators. A frozen twelve-candidate router selected among linear, polynomial, kernel, latent, and structured nonlinear models using separated random data roles for fitting, tuning, ranking, gating, empirical release screening, and final testing.

The positive result is clear within the internal benchmark: the router selects structurally appropriate models and, on the primary forced-expert diagnostic, matches the retrospective test-best candidate on 24/24 tasks. Classical models remain competitive and are selected when their inductive biases fit the generator. Negative controls collapse as expected, and the empirical release screen abstains on difficult tasks.

The negative results are equally important. On the primary benchmark, removing the independent gate or reusing the rank split for release screening changes nothing. A 12-task structural-boundary stress set yields only one gate-induced structural correction and no release-decision difference. When fixed baselines receive the same total observation budget, mean errors still favor MX5 but family-level multiplicity-adjusted significance is not established. The evidence therefore does **not** show that six hard splits are necessary, that the protocol is sample-efficient, or that routing is superior under equal information budgets.

The scientifically defensible conclusion is consequently narrow:

> **Within an internally developed randomized suite of static complex-valued operators, heterogeneous structural hypotheses can be routed accurately, while strict evidence separation appears primarily conservative and is not shown to be a major performance driver when candidate margins are large.**

This is simulation evidence, not external validation. It does not establish performance on real measured systems, temporal dynamical systems, independently authored generator families, or a formally risk-controlled release procedure. The next consequential study should therefore freeze the current router before exposure to an **independently authored benchmark**, use a prospectively specified equal-budget comparison, and, if risk guarantees are desired, evaluate a formally selective release rule on new data rather than retrofitting one to the present results.

Reviewer-Facing Positioning

A suitable venue should evaluate this manuscript as a **simulation-methods study with substantive negative ablations**. The manuscript does not claim a new universal architecture or theorem. Its positive contribution is the demonstration that heterogeneous structural models can be routed accurately inside a controlled synthetic program. Its negative contribution is the finding that the most elaborate evidence-separation design is not shown to improve outcomes on ordinary tasks and provides only modest benefit on a small boundary-stress set.

Accordingly, claims of "state of the art," "general-purpose system identification," "safe deployment," or "formal certification" would be inconsistent with the evidence and are intentionally absent.

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